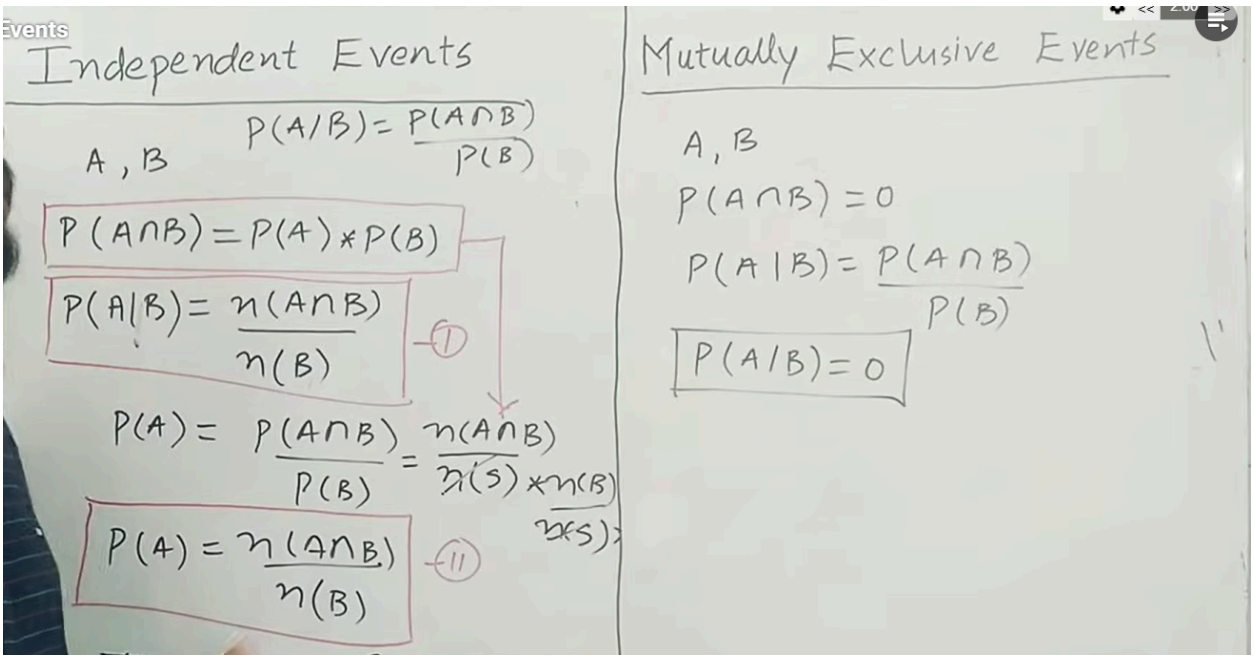
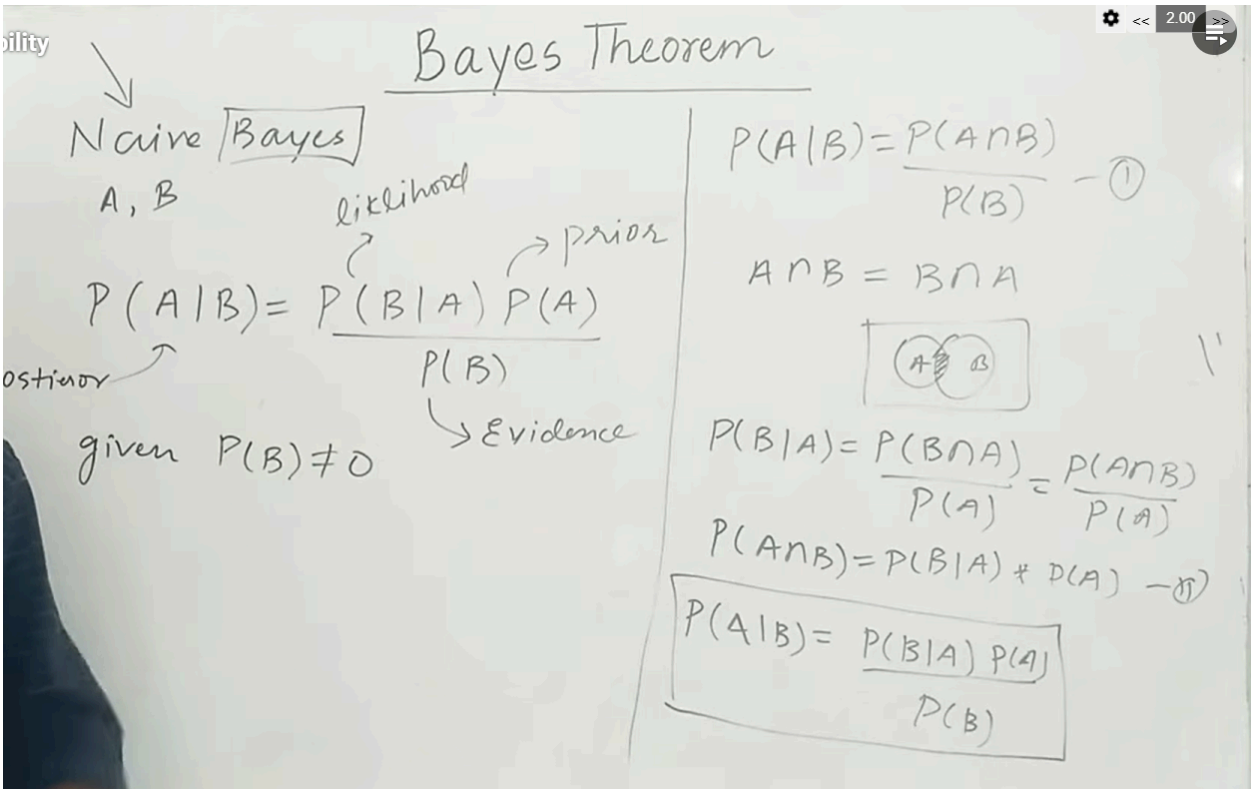
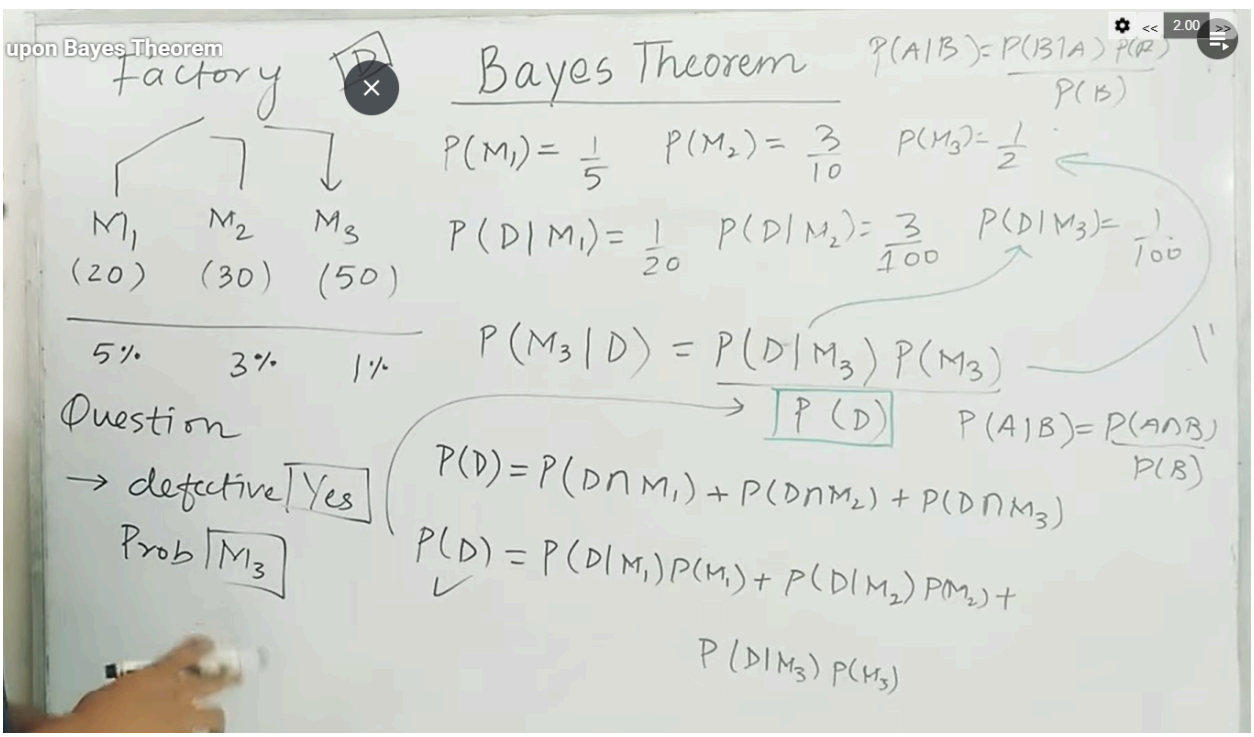
Naive Bayes


Independent Events
A , B
P(A/B) = P(A∩B) / P(B)
P(A∩B) = P(A) * P(B)
P(A|B) = n(A∩B) / n(B)
I
P(A) = P(A∩B) / P(B) = n(A∩B) / n(S) × n(B) / n(S)
P(A) = n(A∩B) / n(B)
II
Mutually Exclusive Events
A , B
P(A∩B) = 0
P(A|B) = P(A∩B) / P(B)
P(A/B) = 0





Better watch the video
https://youtu.be/2PVRG45eVrY?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH

$$X = \{x_1, x_2, x_3 \dots x_n\}$$

$$C_K = \{\boxed{c_1, c_2}, c_3 \dots c_K\}$$

$$P(C_K | X) = P(X | C_K) P(C_K)$$

[Chain Rule for conditional prob]

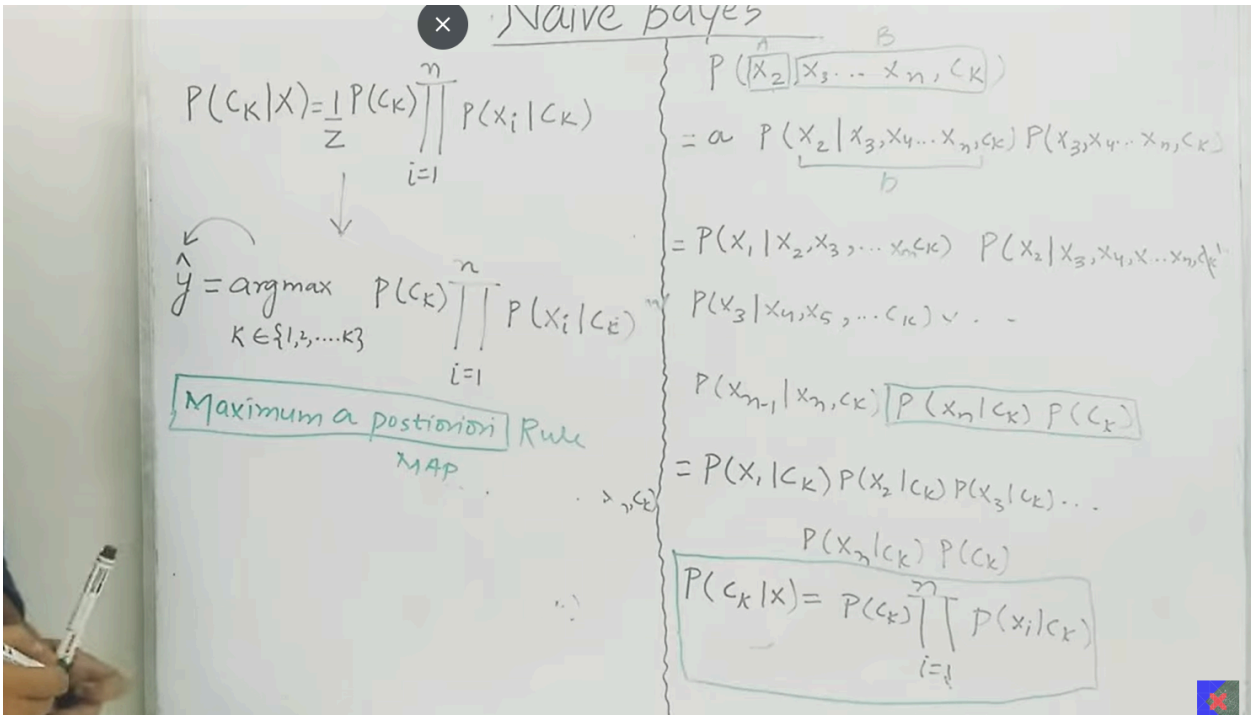
$$P(C_K | X) = P(X, C_K)$$

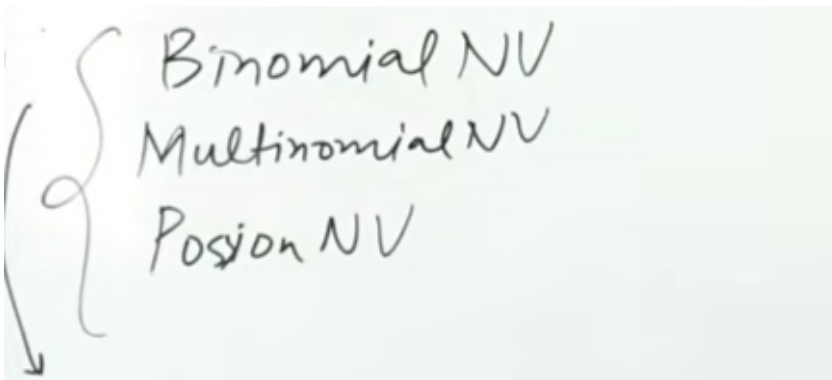
$$A \cap B = P(A|B) P(B)$$

$$= P(\boxed{x_1}, \boxed{x_2, x_3 \dots x_n, C_K})$$

$$= \underbrace{P(x_1 | x_2, x_3 \dots x_n, C_K)}_{a} P(x_2, x_3 \dots x_n, C_K)$$

$$P(C_K | X) = a \times P(x_2, x_3 \dots x_n, C_K)$$


Naive bayes
$P(C_K|X) = \frac{1}{Z} P(C_K) \prod_{i=1}^{n} P(x_i|C_K)$
$\hat{y} = \underset{K \in \{1,2,\dots,K\}}{\operatorname{argmax}} P(C_K) \prod_{i=1}^{n} P(x_i|C_K)$
Maximum a postiori Rule
MAP
$P(\boxed{X_2} \boxed{X_3 \dots X_n, C_K})$
$= a\ P(X_2 | X_3, X_4 \dots X_n, C_K)\ P(X_3, X_4 \dots X_n, C_K)$
$= P(X_1 | X_2, X_3, \dots X_n, C_K)\ P(X_2 | X_3, X_4, X \dots X_n, C_K)$
$P(X_3 | X_4, X_5, \dots C_K) \dots$
$P(X_{n-1} | X_n, C_K)\ \boxed{P(X_n | C_K)\ P(C_K)}$
$= P(X_1 | C_K) P(X_2 | C_K) P(X_3 | C_K) \dots$
$P(X_n | C_K)\ P(C_K)$
$\boxed{P(C_K|X) = P(C_K) \prod_{i=1}^{n} P(x_i|C_K)}$


Binomial NV
Multinomial NV
Posion NV

1️⃣ Naive Bayes Overview

Naive Bayes is a **probabilistic classifier** based on **Bayes' theorem**, assuming that features are **independent given the class**.

Bayes' Theorem

$$P(C \mid X) = \frac{P(X \mid C) \cdot P(C)}{P(X)}$$

Where:

- C = class (e.g., Spam or Ham)
- $X = (x_1, x_2, \ldots, x_n)$ = feature vector
- $P(C)$ = prior probability of class C
- $P(X \mid C)$ = likelihood of features given class C
- $P(C \mid X)$ = posterior probability (what we want)

Naive assumption:

$$P(X \mid C) = P(x_1 \mid C) \cdot P(x_2 \mid C) \cdot \ldots \cdot P(x_n \mid C)$$

2️⃣ Multinomial Naive Bayes (MNB)

Use case:

- **Text classification** with **word counts** (e.g., spam detection, sentiment analysis).
- Handles **discrete features** (like frequency of words).

Formula:

For a document d with words $w_1, w_2, \ldots, w_n$:

$$P(C \mid d) \propto P(C) \cdot \prod_{i=1}^{n} P(w_i \mid C)^{f_i}$$

Where:

- f_i = count of word w_i in the document
- $P(w_i \mid C) = \frac{\text{count of word } w_i \text{ in class } C + 1}{\text{total words in class } C + V}$ (**Laplace smoothing**)
- V = vocabulary size

✅ **Key point:** Takes into account **how many times a word appears**, not just presence/absence.

3️⃣ Bernoulli / Binomial Naive Bayes (BNB)

Use case:

- **Binary features** (word presence/absence).
- Example: "Does the word 'win' appear in this email?" → yes/no.

Formula:

$$P(C \mid X) \propto P(C) \cdot \prod_{i=1}^{n} P(x_i \mid C)^{x_i} \cdot (1 - P(x_i \mid C))^{1-x_i}$$

Where:

- $x_i = 1$ if feature is present, 0 if absent
- P(x_i \mid C) = \frac{\text{# documents with } x_i \text{ in class C} + 1}{\text{total docs in class C} + 2}

✅ **Key point:** Counts **only presence or absence**, not frequency.

4️⃣ Poisson Naive Bayes

Use case:

- Features are **counts that follow a Poisson distribution** (rare events per interval).
- Example: number of emails received per day, number of accidents per intersection, number of clicks per ad.

Formula:

$$P(x_i \mid C) = \frac{\lambda_i^{x_i} e^{-\lambda_i}}{x_i!}$$

Where:

- x_i = observed count
- λ_i = expected count (mean) of feature i in class C

Then, overall probability:

$$P(C \mid X) \propto P(C) \cdot \prod_{i=1}^{n} P(x_i \mid C)$$

✅ **Key point:** Suitable for **rare count data**, unlike multinomial which is more general for text.

5️⃣ Summary Table

Variant	Feature Type	Counts Used?	Common Use Case
Multinomial	Discrete (word counts)	Yes	Text classification, spam, sentiment
Bernoulli/Binomial	Binary (0/1)	No (presence only)	Text classification with binary features
Poisson	Count data	Yes (rare events)	Rare events per interval, event frequency

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6️⃣ Key Notes

- **Laplace smoothing** is important to avoid zero probabilities.
- Choose **multinomial** for word counts, **binomial** for presence/absence, and **poisson** for rare event counts.
- Despite the "naive" assumption, Naive Bayes often performs surprisingly well in practice.

If you want, I can **draw a small diagram showing how Multinomial, Binomial, and Poisson Naive Bayes differ** visually—it makes it super easy to remember.